

Carbon and Its Compounds

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Reference: Coursebook - IIT Foundation Chemistry Class 8; Chapter - Carbon and its Compounds; Page number - 6.1–6.20

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. **Assertion (A):** Diamond is a good thermal conductor.

Reason (R): Diamond is a crystalline allotrope of carbon in which carbon exhibits tetravalency.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** Dry ice is used for cloud seeding.

Reason (R): Dry ice sublimates.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Match the statements of Column A with those of Column B.

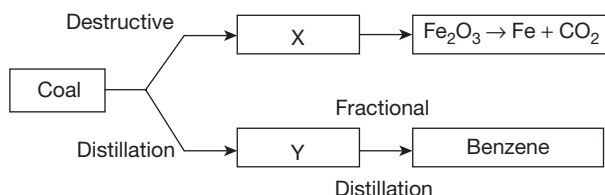
Column A	Column B
(i) Layer lattice structure	(A) Gas carbon
(ii) Spherical cage molecules	(B) Lamp black
(iii) Rigid three-dimensional structure	(C) Graphite
(iv) Amorphous grey solid	(D) Fullerene
(v) Powdery black substance with velvet touch	(E) Diamond

- (a) (i) → (C); (ii) → (A); (iii) → (E); (iv) → (B); (v) → (D)
- (b) (i) → (D); (ii) → (A); (iii) → (C); (iv) → (B); (v) → (D)
- (c) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (B)
- (d) (i) → (E); (ii) → (D); (iii) → (B); (iv) → (C); (v) → (C)

4. Arrange the allotropes in their order of their importance.
- (i) Extraction of yellow phosphorus
 - (ii) Preparation of artificial diamond
 - (iii) Gas masks
 - (iv) Moderator in nuclear reactors
 - (v) Abrasive for cutting glass
- (A) Wood charcoal (B) Bone charcoal
(C) Graphite (D) Diamond
(E) Sugar charcoal
- (a) CDEBA (b) BEACD
(c) CAEDB (d) BCAED
5. The descriptions of some reactions are given below. Arrange the formula of products of the reactions in the order of the descriptions.
- (i) Passing carbon monoxide gas over finely divided iron
 - (ii) Passing carbon dioxide through powdered magnesium
 - (iii) Heating metal oxide (M_2O_3) with carbon monoxide
 - (iv) Passing carbon dioxide through sodium
- (A) M_2CO_3 (B) M
(C) MO (D) $M(CO)_5$
(a) CABD (b) DACB
(c) DCBA (d) CBDA
6. Which of the following reactions does not produce CO_2 ?
- (a) Heating of Na_2CO_3 (b) Reaction of K_2CO_3 with HCl
(c) Heating of $Ca(HCO_3)_2$ (d) Heating of $NaHCO_3$
7. Identify the principle involved in the usage of baking powder in cakes.
- (a) Acid base neutralization
 - (b) Thermal decomposition
 - (c) Dissolution in water
 - (d) Sublimation
8. Which of the following fire extinguishers, when used, involves a physical process instead of a chemical reaction?
- (a) Dry powder extinguisher
 - (b) Foam-type extinguisher
 - (c) Soda-acid extinguisher
 - (d) Liquid CO_2 fire extinguisher
9. Graphite is used for various purposes. Which of the following uses is not related to its layer lattice structure?
- (a) Refractory crucible (b) Pencil lead
(c) Lubricant (d) Both (b) and (c)

Space for rough work

10. Identify the reaction which does not involve the formation of a white precipitate.
- (a) Reaction of carbonic acid with NaOH
 - (b) Reaction of carbonic acid with $\text{Ca}(\text{OH})_2$
 - (c) Reaction of sodium carbonate with MgSO_4
 - (d) Reaction of calcium bicarbonate with NaOH
11. Observe the following flow chart and identify 'X' and 'Y'.



- (a) Coal gas, coal tar
 - (b) Coke, coal gas
 - (c) Coke, coal tar
 - (d) Ammonia, naphthalene
12. If a saturated hydrocarbon has 4 carbon atoms, then the formula of the hydrocarbon could be _____.
- (a) C_4H_6
 - (b) C_4H_8
 - (c) C_4H_{10}
 - (d) C_4H_{12}
13. Identify the name of the harmful gas released in coal mines.
- (a) Carbogen
 - (b) Natural gas
 - (c) Phosgene
 - (d) Marsh gas
14. Which of the following processes is involved in the preparation of carbon black from methane?
- (a) Pyrolysis
 - (b) Carbonization
 - (c) Combustion
 - (d) Catenation
15. Identify the false statement regarding ideal fuel.
- (a) No residue is left over after burning.
 - (b) It has high caloric value.
 - (c) It has high ignition temperature.
 - (d) It burns without producing flames.

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Graphite is a good electrical conductor.

Reason (R): Carbon exhibits a valency of three in graphite.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

2. **Assertion (A):** A sealed bottle containing dry ice explodes after some time.

Reason (R): Dry ice produces very low temperature.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

3. Match the statements of Column A with those of Column B.

Column A (Characteristic Feature)	Column B (Allotrope)
(i) Bucky balls	(A) Coke
(ii) Calcium phosphate is in major proportion	(B) Graphite
(iii) Purest amorphous form of carbon	(C) Diamond
(iv) Hexagonal rings	(D) Animal charcoal
(v) Tetrahedral units	(E) Fullerene

- (a) (i) → (E); (ii) → (A); (iii) → (D); (iv) → (B); (v) → (C)
 (b) (i) → (C); (ii) → (A); (iii) → (D); (iv) → (E); (v) → (B)
 (c) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (E); (v) → (B)
 (d) (i) → (E); (ii) → (D); (iii) → (A); (iv) → (B); (v) → (C)

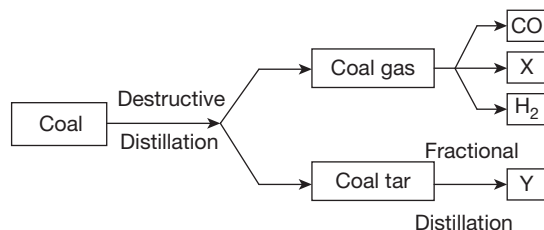
4. Arrange the following allotropes in the order of the raw material used for their preparation, given below.

- (i) Sugar and molten iron (ii) Coke and silicon dioxide
 (iii) Graphite (iv) Coal
 (v) Mustard oil
 (A) Coke (B) Fullerene
 (C) Lamp black (D) Diamond
 (E) Graphite
 (a) DEBAC (b) BECAD
 (c) DABEC (d) BACED

Space for rough work

5. Arrange the products given below in the order of the reactions as follows.
- (i) Neutralization of CO_2 by caustic soda.
 - (ii) Combination of CO_2 and water in the presence of sunlight.
 - (iii) Reaction of CO_2 with Cl_2 gas.
 - (iv) Oxidation of Carbon by CO_2 .
 - (v) CO is passed through blood.
- (A) Lower oxide of carbon (B) Carbohydrate
(C) Carbonate of sodium (D) Carboxy haemoglobin
(E) Phosgene
- (a) CBEDA (b) CBEAD
(c) DCBAE (d) DEBAC
6. Two substances x and y are taken in two test tubes and heated. The gas coming out of only y when passed through lime water turns it milky. Identify x and y .
- (a) ZnCO_3 and MgCO_3 (b) KHCO_3 and MgCO_3
(c) K_2CO_3 and ZnCO_3 (d) Na_2CO_3 and K_2CO_3
7. Under normal conditions, CO_2 is sparingly soluble in water. In which of the following uses of CO_2 , is the above property exploited?
- (a) Preparation of soft drinks
(b) Preservation of food
(c) Usage of mortar to join bricks
(d) Survival of aquatic plants
8. A fire extinguisher has been used for putting off petrol fire. It involves the principle of neutralization. Identify the extinguisher.
- (a) Dry powder extinguisher
(b) Foam-type extinguisher
(c) Liquid CO_2 extinguisher
(d) Soda-acid extinguisher
9. In which application are both layer lattice structure and high melting point of graphite exploited?
- (a) Refractory crucibles (b) Lubricant in machines
(c) Reinforced plastics (d) Moderator in nuclear reactor
10. Which of the following reactions involve both precipitation as well as release of a gaseous product?
- (a) Action of HCl on sodium bicarbonate
(b) Action of HCl on magnesium bicarbonate
(c) Action of HCl on sodium carbonate
(d) Action of HCl on sodium hydroxide

11. Observe the following flow chart and identify 'X' and 'Y'.



- (a) NH_3 , benzene (b) Butane, naphthalene
(c) Methane, ethyl alcohol (d) Methane, naphthalene
12. A saturated hydrocarbon contains 14 hydrogen atoms. What could be the number of carbon atoms in it?
(a) 6 (b) 7
(c) 8 (d) 9
13. Among the following gases, identify the one which is not harmful.
(a) Phosgene (b) Coal gas
(c) Butane (d) Carbon monoxide
14. The name given to the heating of methane in the absence of air is
(a) Carbonization (b) Destructive distillation
(c) Combustion (d) Pyrolysis
15. 'X' and 'Y' are two gaseous fuels and X is found to be a better fuel than Y. Which of the following assumptions regarding X and Y would comply with the above fact?
(i) X has a higher calorific value than Y.
(ii) Y produces ash on combustion at the end.
(iii) X gets ignited at a lower temperature than Y.
(iv) Y catches fire at room temperature whereas X does not.
(a) (i) and (iii) (b) (ii) and (iv)
(c) (i) and (iv) (d) (ii) and (iii)

Space for rough work

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Identify the allotrope of carbon that exists in crystalline form.
(a) Graphite (b) Coal
(c) Coke (d) Lamp black
2. Which of the following is the raw material for the manufacture of graphite in Acheson process?
(a) Silicon carbide (b) Silicon
(c) Silicone (d) Sand
3. In which of the following purposes can diamond be not used?
(a) Tip of deep boring drills
(b) For cutting glasses
(c) As solid lubricant
(d) Generating laser beam in electronics
4. Which of the following can be used to extract metals from their oxides?
(a) Sugar charcoal (b) Bone charcoal
(c) Graphite (d) Fullerene
5. Which of the reactions do not produce CO?
(a) $C + CO_2 \rightarrow$ (b) $H_2C_2O_4 + \text{Conc. } H_2SO_4 \rightarrow$
(c) $HCOOH + \text{Conc. } H_2SO_4 \rightarrow$ (d) $CaCO_3 + HCl \rightarrow$
6. **Assertion (A):** CO is collected in the laboratory by downward displacement of water.
Reason (R): CO is a neutral oxide.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.
7. **Assertion (A):** CO is one of the components of coal gas, water gas and producer gas.
Reason (R): The combustion of CO is highly exothermic.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.
8. Arrange the following fuels in the ascending order of their calorific values.
(A) Hydrogen (B) LPG (C) Petrol (D) Wood
(a) ABCD (b) BCDA (c) CDBA (d) DCBA

9. Arrange the following compounds in the order of their formation during chlorination of methane.
- (A) Chloroform (B) Dichloromethane
(C) Methyl chloride (D) Carbon tetrachloride
(a) BADC (b) CDAB
(c) DBCA (d) CBAD
10. Hydrolysis of calcium carbide liberates ____.
- (a) CO (b) C_2H_2
(c) CO_2 (d) CH_4
11. The reaction given below is associated with ____.
- $$Al_2(SO_4)_3 + 6NaHCO_3 \rightarrow 3Na_2SO_4 + 2Al(OH)_3 + 6CO_2$$
- (a) dry powder fire extinguisher (b) soda-acid fire extinguisher
(c) foam-type extinguisher (d) liquid CO_2 fire extinguisher
12. Which of the following makes lime water milky?
- (a) $Ca(HCO_3)_2$ (b) $CaCO_3$
(c) CaC_2 (d) $NaHCO_3$
13. Which of the following types of coal contains the highest percentage of carbon?
- (a) Anthracite (b) Bituminous
(c) Lignite (d) Peat
14. In which of the following ways is atmospheric CO_2 trapped?
- (a) Respiration (b) Photosynthesis
(c) Consumption of fossil fuel (d) Decomposition of organic matter
15. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Lamp black	(A) Destructive distillation of coal
(ii) Coke	(B) Heating wood saving in absence of air
(iii) Gas carbon	(C) Burning mustard oil in inadequate supply of air
(iv) Wood charcoal	(D) Destructive distillation of animal bones
(v) Bone charcoal	(E) Thermal vaporization of coal followed by condensation

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (E); (iv) $\rightarrow$ (B); (v) $\rightarrow$ (D)
 (b) (i) $\rightarrow$ (A); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (E); (v) $\rightarrow$ (B)
 (c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (E); (iv) $\rightarrow$ (D); (v) $\rightarrow$ (B)
 (d) (i) $\rightarrow$ (A); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (C); (iv) $\rightarrow$ (E); (v) $\rightarrow$ (B)

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. In which of the following are carbon atoms not present in a regular pattern?

- (a) Graphite (b) Diamond
(c) Fullerene (d) Sugar charcoal

2. Which of the following gas is liberated during the manufacture of graphite in Acheson process?

- (a) CO_2 (b) CO
(c) O_2 (d) O_3

3. Which of the following forms of carbon can be used as solid lubricant and why?

- (a) Coal, as it is brittle and high heat resistant.
(b) Fullerene, as it is spherical in shape.
(c) Coke, as it contains less impurities.
(d) Graphite, as it has layered structure and high resistance to heat.

4. Identify the allotrope of carbon used for making electrodes for dry cell.

- (a) Coke (b) Diamond
(c) Wood charcoal (d) Gas carbon

5. Which of the following reactions do not produce CO_2 ?

- (a) Metal carbonate + dilute acid
(b) Metal bicarbonate + dilute acid
(c) Thermal decomposition of lime stone (CaCO_3)
(d) Thermal decomposition of Na_2CO_3

6. **Assertion (A):** CO_2 is collected in the laboratory by upward displacement of air.

Reason (R): CO_2 is a non-supporter of combustion.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

7. **Assertion (A):** CO_2 is extensively used in fire extinguishers.

Reason (R): CO_2 is a good reducing agent.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

8. Arrange the following fuels in the descending order of the ignition temperature.
- (A) Charcoal (B) Petroleum gas
(C) Natural gas (D) Gasoline
(a) DBCA (b) ADBC (c) CDBA (d) DCBA
9. Arrange the following reactants associated with methane in the order of their categories given below.
(Substitution reaction, combustion reaction, cracking and manufacture)
- (A) Heating of CH_4 in absence of air.
(B) Reaction between CH_4 and Cl_2 in the presence of sunlight.
(C) Heating anhydrous sodium acetate with soda lime.
(D) Reaction between CH_4 and O_2 .
(a) CDBA (b) ACBD (c) BDAC (d) BDCA
10. The nature of the product which is not a gas obtained during hydrolysis of calcium carbide is _____.
(a) basic (b) strongly acidic
(c) neutral (d) weakly acidic
11. Thermal decomposition of sodium bicarbonate is associated with _____.
(a) dry powder fire extinguisher
(b) liquid CO_2 fire extinguisher
(c) soda-acid fire extinguisher
(d) foam-type of fire extinguisher
12. Passing excess CO_2 through milky lime water makes it clear due to the formation of _____.
(a) CaCO_3 (b) Na_2CO_3
(c) NaHCO_3 (d) $\text{Ca}(\text{HCO}_3)_2$
13. Which of the following types of coal, on destructive distillation, gives rise to the minimum yield of coke?
(a) Peat (b) Bituminous
(c) Lignite (d) Anthracite
14. Identify the process of releasing CO_2 in the atmosphere.
(a) Dissolution of CO_2 in water bodies
(b) Formation of fossil fuel from dead remains
(c) Respiration
(d) Afforestation